The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 July 2022.

INCH-POUND
MIL-PRF-19500/375K
w/AMENDMENT 1
27 April 2022
SUPERSEDING
MIL-PRF-19500/375K
2 June 2017

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD-EFFECT, N-CHANNEL, DEPLETION MODE, SILICON, TYPES 2N3821, 2N3822, AND 2N3823, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel, junction, silicon, field-effect depletion mode transistors. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device.

1.2 Package outlines. The device package outlines are as follows: TO-72 in accordance with figure 1 and surface mount, UB suffix in accordance with figure 2 for all encapsulated device types.

1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

P_T (1) T_A = +25°C	V_{DG} and V_{DS}			V_{GSR}			I_{GF}	T_J and T_{STG}
	2N3821 2N3821UB	2N3822 2N3822UB	2N3823 2N3823UB	2N3821 2N3821UB	2N3822 2N3822UB	2N3823 2N3823UB		
mW	V dc	V dc	V dc	V dc	V dc	V dc	mA dc	°C
300	50	50	30	50	50	30	10	-55 to +200

(1) Derate linearly, 1.7 mW/°C for $T_A \geq$ +25°C.

1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

	$\lvert y_{fs}\rvert$1 (1) V_{DS} = 15 V dc, f = 1 kHz, V_{GS} = 0			$\lvert y_{fs}\rvert$3 V_{DS} = 15 V dc, V_{GS} = 0			C_{iss} V_{DS} = 15 V dc V_{GS} = 0 f ≤ 1 MHz	C_{rss} V_{DS} = 15 V dc V_{GS} = 0100 kHz ≤ f ≤ 1 MHz		
				f = 100 MHz	100 kHz ≤ f = 100 MHz	f = 200 MHz				
	2N3821 2N3821UB	2N3822 2N3822UB	2N3823 2N3823UB	2N3821 2N3821UB	2N3822 2N3822UB	2N3823 2N3823UB		2N3821 2N3821UB	2N3822 2N3822UB	2N3823 2N3823UB
	µs	µs	µs	µs	µs	µs	pF	pF	pF	pF
Min	1,500	3,000	3,500	1,500	3,000	3,200				
Max	4,500	6,500	6,500				6	3	3	2

See note at end of 1.4.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.4 Primary electrical characteristics - Continued.

	I_{DSS} (1) V_{DS} = 15 V dc V_{GS} = 0			$V_{GS(off)}$ V_{DS} = 15 V dc I_D = 0.5 nA dc			N_F V_{DS} = 15 V dc V_{GS} = 0, R_G = 1 MΩ f = 10 Hz		V_{DS} = 15 V dc V_{GS} = 0, R_G = 1 MΩ f = 1 kHz	V_{DS} = 15 V dc V_{GS} = 0, R_G = 1 MΩ f = 105 MHz
	2N3821 2N3821UB	2N3822 2N3822UB	2N3823 2N3823UB	2N3821 2N3821UB	2N3822 2N3822UB	2N3823 2N3823UB	2N3821 2N3821UB	2N3822 2N3822UB	All types	2N3823 2N3823UB
	mA dc	mA dc	mA dc	V dc	V dc	V dc	dB		dB	dB
Min	0.5	2.0	4.0							
Max	2.5	10.0	20.0	4.0	6.0	8.0	5.0		2.5	2.5

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, “JANTXV” and "JANS".

1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

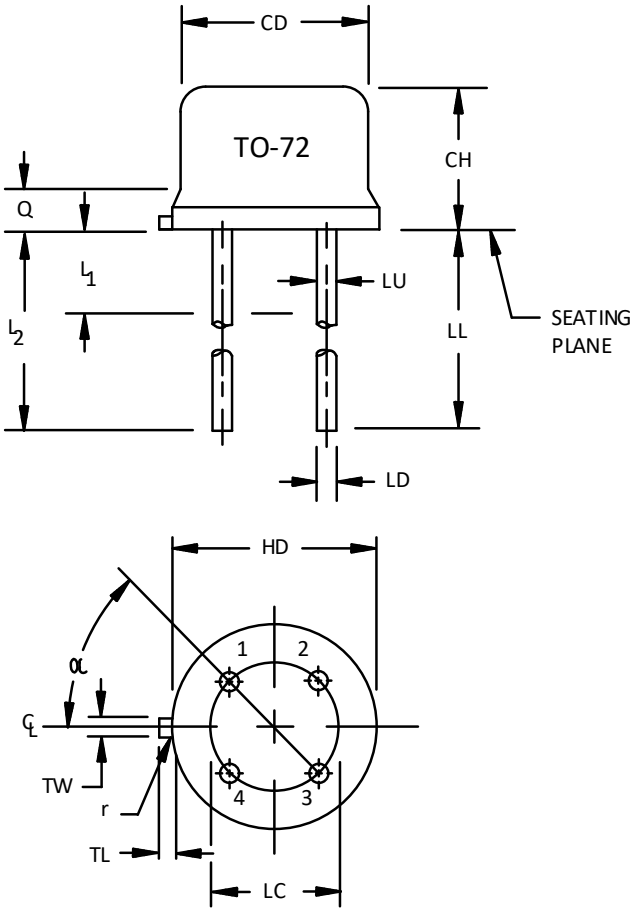
1.5.2.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "3821", "3822", and "3823".

1.5.2.3 Suffix letters. No suffix letters are used on devices that are packaged in the TO-72 package of figure 1. The suffix letters "UB" are used on devices that are packaged in the surface mount package of figure 2.

1.5.3 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

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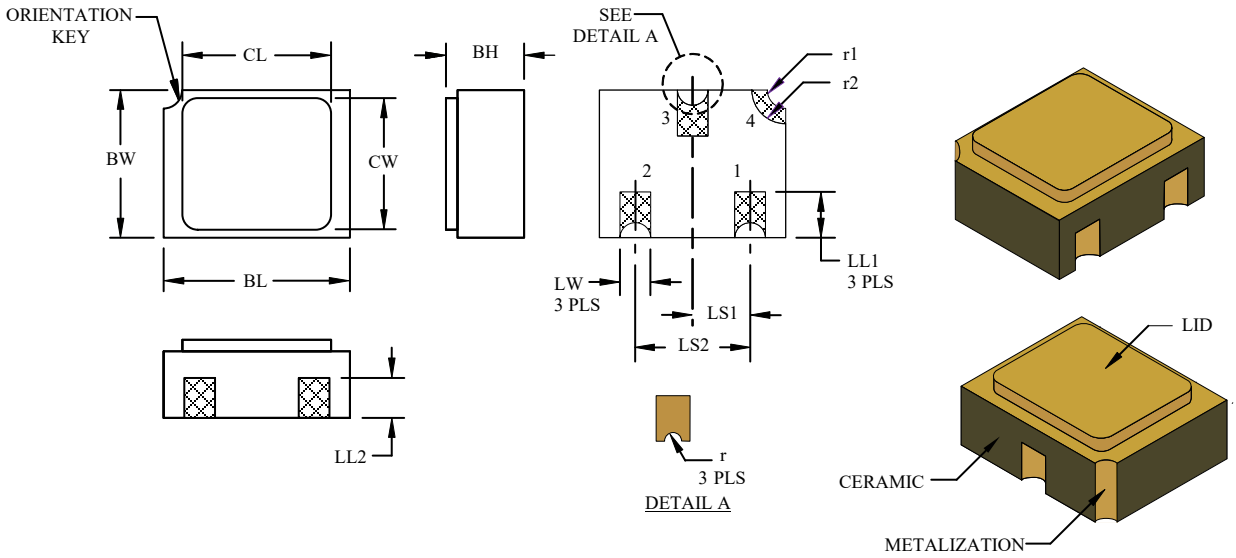
Ltr	Inches	Inches	mm	mm	Notes
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
L1		.050		1.27	
L2	.250		6.35		
LC	.100 TP		2.54 TP		
LD	.016	.021	0.41	0.53	2, 6
LL	.500	.750	12.70	19.05	6
LU	.016	.019	0.41	0.48	3, 6
Q		.040		1.02	
r		.007		0.18	
TL	.028	.048	0.71	1.22	8
TW	.036	.046	0.91	1.17	
α	45° TP				



NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. When measured in a gauging plane .054 +.001, -.000 (1.37 +.03, -.00 mm) before the seating plane of the transistor, maximum diameter leads shall be within .007 (.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance.
5. The active elements are electrically insulated from the case.
6. All 4 leads.
7. Lead 1 is the source, lead 2 is the drain, lead 3 is the gate, and lead 4 is the case.
8. Symbol TL is measured from HD maximum.
9. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (similar to TO-72).

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*

Symbol	Inches	Inches	mm	mm	Note
	Min	Max	Min	Max	
BH	.046	.056	1.17	1.42	
BL	.115	.128	2.92	3.25	
BW	.085	.108	2.16	2.74	
CL		.128		3.25	
CW		.108		2.74	
LL1	.022	.038	0.56	0.96	
LL2	.017	.035	0.43	0.89	
LS1	.036	.040	0.91	1.02	
LS2	.071	.079	1.81	2.01	
LW	.016	.024	0.41	0.61	
r		.008		.203	
r1		.012		.305	
r2		.022		.559	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas.
4. Lid material: Kovar.
5. Pad 1 = Drain, Pad 2 = Source, Pad 3 = Gate, Pad 4 = Shielding connected to the lid.

* FIGURE 2. Physical dimensions, surface mount (2N3821UB, 2N3822UB, AND 2N3823UB).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

R_{ISO} ..... Insulation resistance, gate lead to case.
S ..... Unit of conductance or admittance (formerly mho).
V_1 ..... rms value of the specified reference voltage in the specified circuit.
V_2 ..... rms value of the measured voltage between the specified points in the specified circuit.
$|y_{os}|$ ..... Magnitude of small-signal common-source short-circuit output admittance.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (TO-72) and figure 2 (surface mount, UB) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

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3.7 Marking. Marking shall be in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturers symbol or logo. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix and the "UB" suffix can also be omitted.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4 VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

*

Screen	Measurements for JANS level	Measurements for JANTX and JANTXV levels
11	I_{GSSR1}, I_{DSS}, and $\|y_{fs}\|1$	I_{GSSR1}, I_{DSS}, and $\|y_{fs}\|1$
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{DSS} = ±10 percent of initial value; $\Delta\|y_{fs}\|1$ = ±20 percent initial value.	Subgroup 2 of table I herein; ΔI_{DSS} = ±10 percent of initial value; $\Delta\|y_{fs}\|1$ = ±20 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: 2N3821, 2N3821UB, 2N3822, and 2N3822UB: V_{GS} = 40 V dc; V_{DS} = 0; T_A = +150°C. 2N3823, 2N3823UB: V_{GS} = 24 V dc; V_{DS} = 0; T_A = +150°C. NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables E-VIA (JANS) and E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and 4.4.2.1 and 4.4.2.2 herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	2N3821, 2N3821UB, 2N3822, and 2N3822UB: V_{DS} = 30 V dc, V_{GS} = 0; 2N3823, 2N3823UB: V_{DS} = 15 V dc, V_{GS} = 0; at T_A = +30°C ±5°C. t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on the devices shall be permitted.
B5	1027	V_{DS} = 0; 2N3821, 2N3821UB, 2N3822, and 2N3822UB: V_{GS} = 40 V dc; 2N3823, 2N3823UB: V_{GS} = 24 V dc; at T_A = +225°C for 168 hours or T_A = +200°C for 260 hours.

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1027	2N3821, 2N3821UB, 2N3822, 2N3822UB, and 2N3822: V_{GS} = 40 V dc, V_{DS} = 0; 2N3823, 2N3823UB: V_{DS} = 0, V_{GS} = 24; at T_A = +150°C ±5°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Delta measurements shall be in accordance with table II herein.

Subgroup	Method	Condition
C2	2036	Test condition E (not applicable to "UB" suffix version).
C6	1026	2N3821, 2N3821UB, 2N3822, 2N3822UB, and 2N3822: V_{GS} = 40 V dc, V_{DS} = 0; 2N3823, 2N3823UB: V_{DS} = 0, V_{GS} = 24; at T_A = +150°C ±5°C.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Delta measurements shall be in accordance with table II herein.

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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Disposition of case lead during electrical measurements. All electrical measurements (except for case insulation test) operating life test shall be performed with the case lead connected to the source.

4.5.2.1 Case insulation resistance test. Case insulation resistance test shall be measured between the case and gate lead.

4.5.3 Small-signal common-source short-circuit input (or output) conductance. These tests shall be conducted with a General Radio transfer function and admittance bridge model 1607A (or suitable equivalent) in accordance with the portion of its accompanying handbook which is applicable to this measurement. A Hewlett-Packard generator model 608D and a Nems-Clarke receiver model 1502A (or suitable equivalents) shall be used with the transfer function and admittance bridge.

4.5.4 10 Hz and 1 kHz spot noise figure tests. These tests shall be conducted with a model 2173C Quan Tech Laboratories test set, or equivalent. Conditions shall be as specified in table I.

4.5.5 105 MHz spot noise figure test. This test shall be conducted with the equipment and circuit shown on figures 3 and 4, or suitable equivalent.

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TABLE I. Group A inspection .

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Gate-source breakdown voltage	3401	Bias condition C; I_G = 1.0 µA dc; V_{DS} = 0	$V_{(BR)GSSR}$			
2N3821, 2N3821UB				50		V dc
2N3822, 2N3822UB				50		V dc
2N3823, 2N3823UB				30		V dc
Gate reverse current	3411	Bias condition C;	I_{GSSR1}			
2N3821, 2N3821UB		V_{GS} = 30 V dc; V_{DS} = 0			0.1	nA dc
2N3822, 2N3822UB		V_{GS} = 30 V dc; V_{DS} = 0			0.1	nA dc
2N3823, 2N3823UB		V_{GS} = 20 V dc; V_{DS} = 0			0.5	nA dc
Zero-gate-voltage drain current	3413	Bias condition C; V_{DS} = 15 V dc; V_{GS} = 0; pulsed (see 4.5.1)	I_{DSS}			
2N3821, 2N3821UB				0.5	2.5	mA dc
2N3822, 2N3822UB				2.0	10.0	mA dc
2N3823, 2N3823UB				4.0	20.0	mA dc
Gate-source voltage	3403	V_{DS} = 15 V dc;	V_{GS}			
2N3821, 2N3821UB		I_D = 50 µA dc		0.5	2.0	V dc
2N3822, 2N3822UB		I_D = 200 µA dc		1.0	4.0	V dc
2N3823, 2N3823UB		I_D = 400 µA dc		1.0	7.5	V dc
Gate-source cutoff voltage	3403	V_{DS} = 15 V dc; I_D = 0.5 nA dc	$V_{GS(off)}$			
2N3821, 2N3821UB					4.0	V dc
2N3822, 2N3822UB					6.0	V dc
2N3823, 2N3823UB					8.0	V dc
Case insulation resistance	1016	Test condition A; (see 4.5.2.1)	R_{ISO}	10^{10}		Ω
Small-signal, common-source, short-circuit, forward transadmittance	3455	V_{DS} = 15 V dc; V_{GS} = 0; f = 1 kHz; pulsed (see 4.5.1)	$\|y_{fs}\|1$			
2N3821, 2N3821UB				1,500	4,500	µS
2N3822, 2N3822UB				3,000	6,500	µS
2N3823, 2N3823UB				3,500	6,500	µS
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Gate reverse current	3411	Bias condition C;	I_{GSSR2}			
2N3821, 2N3821UB		V_{GS} = 30 V dc; V_{DS} = 0			0.1	µA dc
2N3822, 2N3822UB		V_{GS} = 30 V dc; V_{DS} = 0			0.1	µA dc
2N3823, 2N3823UB		V_{GS} = 20 V dc; V_{DS} = 0			0.5	µA dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Low-temperature operation:		T_A = -55°C				
Small-signal, common-source, short-circuit, forward transadmittance	3455	V_{DS} = 15 V dc; V_{GS} = 0; f = 1 kHz; pulsed (see 4.5.1)	$\|y_{fs}\|2$			
2N3821, 2N3821UB					6,750	µS
2N3822, 2N3822UB					9,750	µS
2N3823, 2N3823UB					9,750	µS
Subgroup 4						
Small-signal common-source, short-circuit, output admittance	3453	V_{DS} = 15 V dc; V_{GS} = 0; f = 1 kHz; pulsed (see 4.5.1)	$\|y_{os}\|$			
2N3821, 2N3821UB					10	µS
2N3822, 2N3822UB					20	µS
2N3823, 2N3823UB					35	µS
Small-signal common-source, short-circuit, input capacitance	3431	V_{DS} = 15 V dc; V_{GS} = 0; 100 kHz ≤ f ≤ 1 MHz	C_{iss}		6	pF
Small-signal common-source, short-circuit, reverse-transfer capacitance	3453	V_{DS} = 15 V dc; V_{GS} = 0; 100 kHz ≤ f ≤ 1 MHz	C_{rss}			
2N3821, 2N3821UB, 2N3822, 2N3822UB					3	pF
2N3823, 2N3823UB					2	pF
Small-signal common-source, short-circuit, forward transadmittance	3455	V_{DS} = 15 V dc; V_{GS} = 0	$\|y_{fs}\|3$			
2N3821, 2N3821UB		f = 100 MHz		1,500		µS
2N3822, 2N3822UB		f = 100 MHz		3,000		µS
2N3823, 2N3823UB		f = 200 MHz		3,200		µS
Small-signal common-source, short-circuit, input conductance		V_{DS} = 15 V dc; V_{GS} = 0; f = 200 MHz; (see 4.5.3)	g_{is}			
2N3823 and 2N3823UB only					800	µS

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Small-signal common-source, short-circuit, output conductance		V_{DS} = 15 V dc; V_{GS} = 0; f = 200 MHz; (see 4.5.3)	g_{os}			
2N3823 and 2N3823UB only					200	µS
Common-source spot noise figure		V_{DS} = 15 V dc; V_{GS} = 0; R_G = 1 MΩ; (see 4.5.4)	NF_1			
2N3821, 2N3821UB, and 2N3822, 2N3822UB only		f = 10 Hz			5.0	dB
All types		f = 1 kHz			2.5	dB
Common-source spot noise figure		V_{DS} = 15 V dc; V_{GS} = 0; R_G = 1 kΩ; f = 105 MHz; (see 4.5.5)	NF_2			
2N3823, 2N3823UB only					2.5	dB
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups B, C, and E delta measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1.	Gate-source voltage 2N3821, 2N3821UB 2N3822, 2N3822UB 2N3823, 2N3823UB	3403	V_{DS} = 15 V dc I_D = 50 µA dc I_D = 200 µA dc I_D = 400 µA dc	ΔV_{GS}	±50 mV change from previously measured value	
2.	Zero-gate-voltage drain current	3413	Bias condition C; V_{DS} = 15 V dc; V_{GS} = 0; pulsed (see 4.5.1)	ΔI_{DSS}	±15 percent change from previously measured value.	
3.	Small-signal, common-source short-circuit, forward transadmittance	3455	V_{DS} = 15 V dc; V_{GS} = 0; f = 1 kHz; pulsed (see 4.5.1)	$\Delta\|y_{fs}\|1$	±25 percent change from previously measured value.	

1/ The delta measurements for table E-VIA (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table II herein, steps 2 and 3.
   b. Subgroup 4, see table II herein, steps 1, 2, and 3.
   c. Subgroup 5, see table II herein, steps 2 and 3.

2/ The delta measurements for table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:
   Subgroup 3, see table II herein, step 3.

3/ The delta measurements for table E-VII of MIL-PRF-19500 are as follows:
   a. Subgroups 2 and 3, table II herein, steps 1, 2, and 3 (for JANS only).
   b. Subgroup 6, see table II herein, steps 1, 2, and 3, (for JANS) and step 3 (for JAN, JANTX, and JANTXV).

4/ The delta measurements for table E-IX of MIL-PRF-19500 are: Subgroups 1 and 2, see table II herein, all steps.

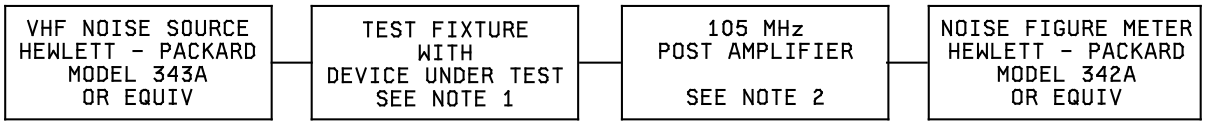
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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	-55°C to +150°C, 500 cycles	
Hermetic seal Fine leak Gross leak	1071	As applicable	
Electrical measurements		See table I, subgroup 2 and table II, all steps	
Subgroup 2 1/			45 devices c = 0
Steady-state operating life	1026	V_{DS} = 0; 2N3821, 2N3821UB, 2N3822, and 2N3822UB: V_{GS} = 40 V dc; 2N3823, 2N3823UB: V_{GS} = 24 V dc; at T_A = +225°C for 168 hours or T_A = +200°C for 260 hours.	
Electrical measurements		See table I, subgroup 2 and table II, all steps	
Subgroups 4 and 5			
Not applicable			
Subgroup 7			45 devices c = 0
Reverse stability	1033	Condition A	

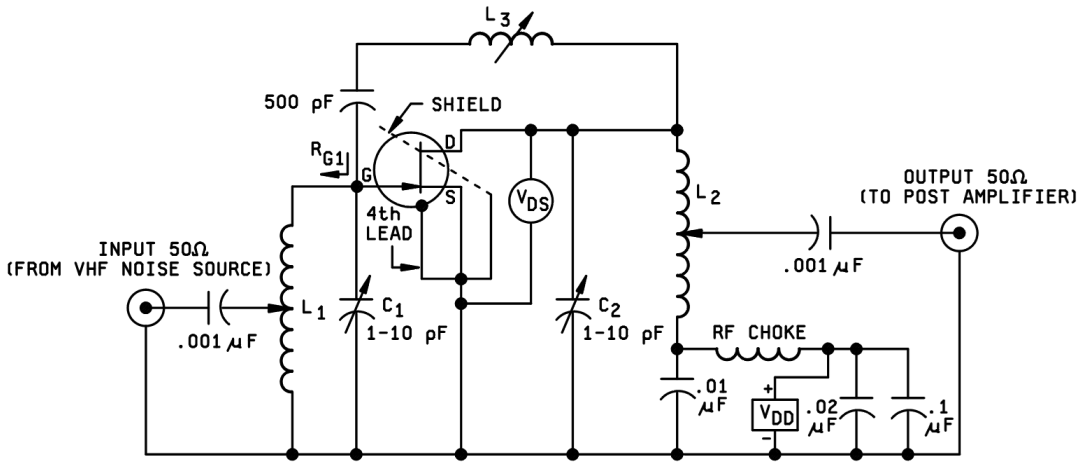
1/ A separate sample for each test shall be pulled.

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NOTES:
1. The test fixture and procedure shall be in accordance with figure 4.
2. The post amplifier shall contain suitable biasing circuits and shall be constructed utilizing very high frequency design technique.

FIGURE 3. Block diagram for 105 MHz spot noise figure test.



CIRCUIT-COMPONENT INFORMATION:

L_1: Approximately 8.5 turns of No. 16 tinned copper; diameter approximately .375 inch (9.52 mm); length approximately .9 inch (22.8 mm); tapped at approximately 2.5 turns (adjust to give R_{G1} = 100 Ω); parallel resistance = 40 kΩ; tunes at approximately 8 pF.

L_2: Approximately 13.5 turns of No. 16 tinned copper; diameter approximately .375 inch; length approximately 1.2 inches (30.5 mm); tapped at approximately 5 turns; parallel resistance = 40 kΩ; tunes at approximately 4 pF.

L_3: Approximately 15 turns of No. 20 tinned copper.

TEST PROCEDURE:

1. Neutralization shall be accomplished as follows:
   a. Insert a typical device from the units to be tested.
   b. Using a sweep generator and an oscilloscope adjust the input (C_1) and output (C_2) for maximum gain at 105 MHz.
   c. Reverse the input and output of the test jig and tune L_3 for a null.
   d. Return the input and output of the test jig to normal.
2. Once neutralization has been obtained, the devices may be tested by inserting them in the test jig and adjusting C_1 and C_2 to obtain minimum noise figure for each device.

FIGURE 4. 105 MHz spot noise figure test fixture.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

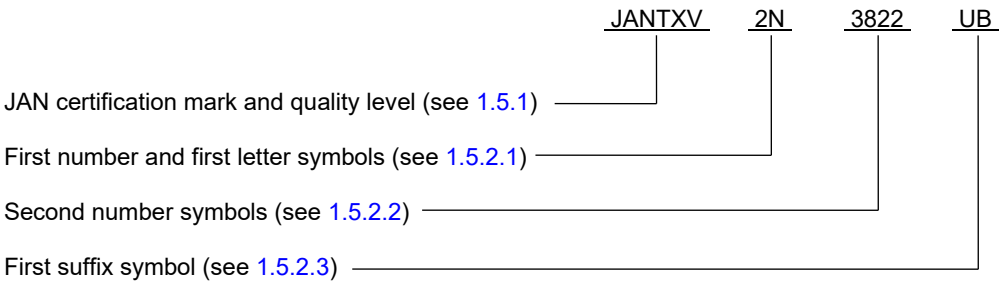
c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QPDSIS-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.5 List of PINs.

6.5.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN2N3821	JANTX2N3821	JANTXV2N3821	JANS2N3821
JAN2N3822	JANTX2N3822	JANTXV2N3822	JANS2N3822
JAN2N3823	JANTX2N3823	JANTXV2N3823	JANS2N3823

6.6 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-6939 or DSN 850-6939.

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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